

Predicting Student Final Grades (G3)

A comparative study of Linear Regression, k-NN, and Decision Tree | UCI Student Performance Dataset

1. Introduction

This report investigates the prediction of students' final mathematics grades (G3) using three regression models: Linear Regression, k-NN, and Decision Tree. G1 and G2 are excluded to reflect a scenario where only demographic and social information is available. Performance is evaluated using MAE and RMSE on a held-out test set. Section 7 also discusses Random Forest Regressor as a possible alternative model.

2. Description of the dataset

The analysis uses the UCI Student Performance dataset (Cortez & Silva, 2008), specifically the mathematics subset. The dataset contains 395 students from two Portuguese schools and records information on demographic, family, school-related, social, and academic factors, including G1, G2, and G3, across 33 variables.

3. Data preparation

To keep the task practically meaningful, G1 and G2 were removed from the predictors, and 30 input variables were left before encoding. One-hot encoding was applied to 17 categorical variables, expanding the feature matrix to 56 columns. No missing values were found. The data were split 80/20 into training and test sets (`random_state = 42`), and `StandardScaler` was applied only for k-NN because it is distance-based.

4. Exploratory analysis

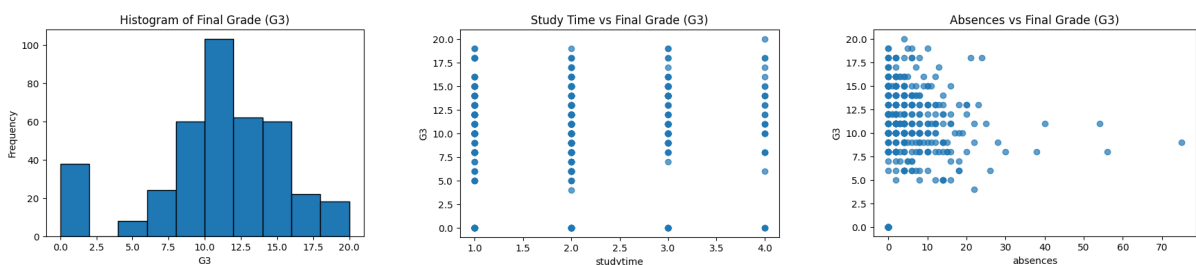


Figure 1. Left: Distribution of G3. Centre: Study time vs G3. Right: Absences vs G3.

The histogram of G3 shows that most grades are concentrated in the middle range, with a noticeable group of 38 students at G3=0. The scatter plots suggest a weak positive relationship between study time and G3, and a mild negative relationship between absences and G3. However, both patterns are fairly noisy, confirming that no single predictor dominates the prediction task.

5. The three required models

In this project, Linear Regression was used as a simple baseline model, predicting G3 as a linear combination of the encoded features. k-Nearest Neighbors (k-NN) Regressor predicted G3 by averaging the grades of the 5 nearest training samples, with feature scaling applied because the method is distance-based. Decision Tree Regressor predicts G3 by recursively splitting the feature space into smaller regions. A constant prediction is then assigned within each final region, and the maximum depth was limited to 5 to reduce overfitting.

6. Results and comparison

Model	MAE	RMSE
Linear Regression	3.395✓	4.196✓
k-NN Regressor(k=5)	3.754	4.621
Decision Tree(max_depth = 5)	3.536	4.621

Table 1. Test-set MAE and RMSE (lower is better; ✓ marks the best value per metric).

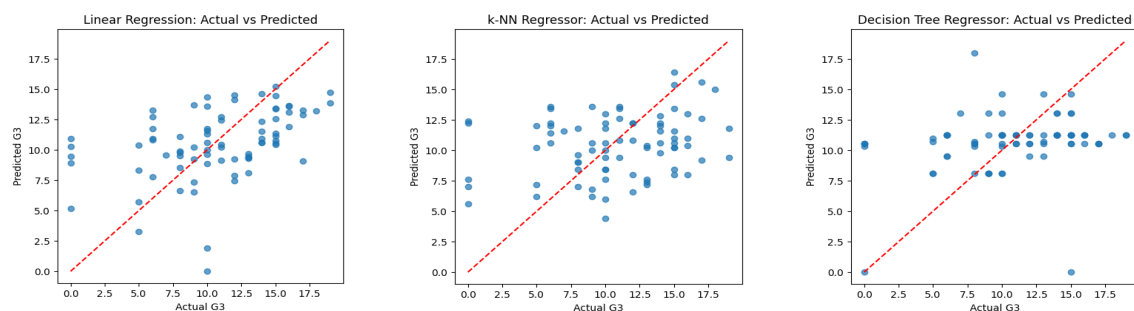


Figure 3. Actual vs Predicted G3: Linear Regression (left), k-NN (center), Decision Tree (right). Red dashed line = perfect prediction.

6.1 Which model performed best?

Among the three required models, Linear Regression performed best, achieving the lowest MAE (3.395) and RMSE (4.196). In contrast, k-NN Regressor produced the highest MAE, while Decision Tree Regressor had a slightly lower MAE than k-NN but the same RMSE (4.621). The consistent gap between RMSE and MAE across all models — roughly 0.8–1.1 grade points — indicates the presence of several large individual errors, most likely caused by the 38 students with G3 = 0, whose extreme outcomes all models struggled to predict accurately from demographic and social features alone.

6.2 Did the linear model perform surprisingly well?

Linear Regression performed relatively well compared with the other two models, which may seem surprising given that the exploratory plots suggested noisy rather than strong linear relationships. However, after one-hot encoding, the model used 56 features, and many small additive effects may have been captured effectively at a global level. Figure 3 also shows that predictions followed the general trend in G3, though compressed toward the middle range – with low grades overestimated and high grades underestimated – and a few large individual errors remained, indicating the model still struggled on some specific students.

6.3 Was k-NN sensitive to preprocessing?

Yes, k-NN was clearly sensitive to preprocessing. Since k-NN is based on Euclidean distance, scaling was necessary to prevent variables with larger numeric ranges from dominating the distance calculation. Even after scaling, however, k-NN performed worst overall. One possible reason is that the encoded dataset has high dimensionality, which makes the notion of 'nearest neighbors' less informative, causing very low grades to be pulled upward toward the center of the distribution.

6.4 Did the decision tree appear more flexible?

The Decision Tree appeared more flexible than Linear Regression in Figure 3 because its predictions were distributed across several discrete levels rather than following a single smooth trend, suggesting the model was adapting to different local regions of the data. However, this flexibility was piecewise rather than continuous: the tree assigns a single constant prediction to all samples within each leaf node, which limits its ability to capture finer variation within each region and may help explain why high grades were often underestimated. As a result, this apparent flexibility did not translate into better overall performance, since its RMSE remained higher than that of Linear Regression, and the plot still revealed a few severe individual errors, including one case where an actual grade of around 15 was predicted close to 0.

6.5 Which model would be recommended for this dataset?

Based on these results, Linear Regression would be the recommended model among the three required methods. It achieved the best results on both evaluation metrics and followed the overall trend in the data more effectively. In addition, it is simple, computationally efficient, and easy to interpret. However, the overall errors remain non-trivial, so even the best-performing model still has limited predictive accuracy on this dataset.

7. Alternative Model Discussion: Random Forest Regressor

7.1 Why might this model be suitable for this dataset?

A Random Forest Regressor may be suitable for this dataset because the earlier results suggest that the relationship between the predictors and G3 is noisy and may not be fully captured by a single global linear pattern. For example, the exploratory plots show only a weak positive association between study time and G3, and a mild negative association between absences and G3, while the grades remain widely scattered at the same predictor levels. This suggests that individual predictors have limited explanatory power on their own. At the same time, the Decision Tree showed some local flexibility, indicating that nonlinear structure and interactions between variables may exist in the data. As an ensemble of many decision trees, Random Forest can model nonlinear relationships through recursive regional splits and can naturally capture interactions through sequential splitting. In addition, by averaging predictions across many trees, it reduces the instability of a single Decision Tree and is therefore usually more robust.

7.2 What preprocessing or assumptions might it involve?

Random Forest would use the same prepared predictors as the main models, including the removal of G1 and G2 and one-hot encoding of categorical variables. Unlike k-NN, it does not rely on Euclidean distance, so feature scaling is not essential. It also makes fewer explicit assumptions about the

functional form of the relationship between predictors and G3, since it can model complex patterns through an ensemble of trees rather than one fixed equation.

7.3 What are its main strengths?

A major strength of Random Forest is that it is better suited than Linear Regression to the nonlinear and locally varying patterns that may exist in this problem. Linear Regression uses one global equation for all students, and therefore assumes that the effect of a given predictor is fixed across the whole dataset. By contrast, Random Forest can first divide students into smaller groups based on key characteristics, and then learn more suitable prediction rules within each group. For example, the same increase in study time may be more beneficial for students with low absences, but less effective for students with many past failures. In addition, because Random Forest combines the predictions of many trees rather than relying on the split path of a single tree, it is usually more stable than one Decision Tree.

7.4 What are its main weaknesses?

However, Random Forest also has clear limitations. First, it is less interpretable than Linear Regression, because its predictions are produced by combining the results of many trees rather than by a single, easily understood formula. Second, the dataset is relatively small, with only 395 students but 56 predictors after one-hot encoding. As the model divides students into smaller groups based on their characteristics, the number of useful examples within each group may become limited, so clear local patterns are not always easy to learn. In such cases, averaging across many trees can pull predictions back toward the more common middle score range rather than producing very low or very high values. As a result, low grades may be overestimated and high grades underestimated. Finally, the $G3=0$ subgroup contains only 38 students. If these cases reflect absence or other special circumstances rather than ordinary low performance, Random Forest may still struggle to distinguish them reliably using the available background information.

7.5 Would it be expected to perform better or worse than the three required models?

Overall, Random Forest would be expected to perform better than the single Decision Tree, because it keeps the flexibility of tree-based modelling while reducing variance through averaging. It would also likely outperform k-NN, since it is less affected by the high-dimensional encoded feature space. Compared with Linear Regression, it might perform better if nonlinear relationships and feature interactions are important, although this cannot be guaranteed. Therefore, Random Forest appears to be a plausible alternative model for this dataset.

8. Conclusion

Among the three required models, Linear Regression performed best, achieving the lowest MAE and RMSE on the test set. This suggests that, despite the noisy and partly nonlinear relationships in the data, a global linear model was still able to capture the overall pattern more effectively than k-NN and Decision Tree. It is therefore the most appropriate of the three required methods, as it offered the best balance of predictive accuracy, simplicity, and interpretability. The discussion of Random Forest also suggests that more flexible ensemble models may be worth exploring when nonlinear effects and interactions are important. One limitation of this investigation is the relatively small dataset size, which means the results may be sensitive to the particular train-test split.